the damage would still be catastrophic, it's like the difference between getting shot with a bullet from a sniper rifle or the shot from a shotgun, the effect would ultimately be the same.

The problem we run into here is how to deal with these asteroids. An asteroid that is a single solid chunk would be comparatively easy to deal with, but a debris cloud would just absorb the impact of kinetic projectiles and nuclear weapons (we'll come to that later) and regroup again if scattered. Such imaginative ideas as solar sails won't work and we would have to try something as exotic as gravity tethers. Another problem is that large asteroids usually have a couple of mini-moons in tow, meaning that an impact would involve not one, but maybe 2 or even 3 separate asteroids.

Another aspect to consider is that while an asteroid might seem to be very hard, many are actually hollow and may even have vast quantities of ice, frozen just beneath the surface. The continuous venting of such ice adds to the complexity when it comes to tracking and predicting the orbit of such bodies.

Solutions

There has been no shortage of ideas when it comes to trying to stop an incoming asteroid, not to mention dozens of movies and an even larger number of books. NASA and many other space agencies have even conducted detailed studies and have even published papers on the subject. There are primarily two things you do to an asteroid, destroy it or deflect it and what we choose to do will depend on the size and composition of the asteroid.

A debris field trailing a newly forming planet

A militarized moon

The moon is one of the best hopes we have to stop an incoming asteroid. The advantage that the moon has over the Earth is that it has a non-existent atmosphere, meaning that rockets (missiles) can be launched at a fraction of the fuel cost of a launch from Earth, and that the gravity is so low, that we can actually build a massive mass driver, a sort of electromagnetic rail gun, that can hurl massive chunks of rock into space at incredible speeds. The reason we need this is because if we have to slow down or destroy a massive asteroid, we need a launch platform for the launch of projectiles at a massive scale if we even hope to do anything to an asteroid. Most asteroids, the large ones, are moving so fast and have so much momentum, that even a few 100 nuclear missiles might just barely push it off course by a fraction of a degree; that fraction of a degree will be all that will be needed though, given sufficient warning.

With debris cluster asteroids, kinetic impacts are completely ruled out and nuclear missiles have limited use. Unlike on Earth, the actual shockwave from a nuclear blast would be minimal because there is nothing to propagate it. That said, the sheer heat of the blast might serve very well in vapourising portions of the asteroid and given enough time and a sufficient number of missiles, we might actually succeed in reducing its size to a bare minimum. A more feasible, but long-term plan, involves the placement of a sufficiently large spacecraft or artificial moon near enough to such an asteroid that the gravitational attraction between the two will be enough to change the trajectory of the cluster. A gravity tether if you will.

So can we survive?

Yes, we can. We would need the cooperation of every single country on the Earth, their stockpile of missiles, booster rockets, knowledge and skill, but yes, we can survive an imminent collision if we can detect it in time. If we don't detect the asteroid in time (which is actually quite possible)? Well, let's just hope it's a large asteroid, then we won't even be aware that something is wrong and won't care either way!

Asteroids: The source of all life?

It has long been surmised that asteroids were the source of most of the water on our planet. It might very well be true that the impact of massive asteroids brought water to the surface of the Earth, but how would that be possible? As mentioned in the article, asteroids are not what you may think they are. Many of them are possibly vast, hollow, honeycombed structures encasing enormous quantites of frozen water and/or other organic compunds. Multiple collisions (with the Earth and other planets) over the course of millions of millennia would have deposited these vast quantites of water on the planets. Most planets when formed, were literally molten balls of fire. It is hard to imagine the immense amount of fricrtion and pressure that was involved at the time planets form, but it was hot enough to completely vapourize all water from the surface of most planets. Asteroids might prove to be the answer as to how water actually made its way back to planets such as ours. Planets that managed to develop an atmosphere over time, sufficient enough to trap the vaporizing water and prevent it from escaping into the vast depths of space.